Azure Local: Troubleshooting Guide

Common deployment, Day-2 and fault symptoms with a likely cause and the action to take. Start with the cheapest read-only check, and change one thing at a time.

Status: prepared 2026-10-06 against Microsoft Learn (2609 release family). Where a cause depends on your release or hardware, the text says so. This guide does not replace support: collect diagnostics and escalate when the action column does not resolve the symptom.

First checks

```
Get-ClusterNode
Get-PhysicalDisk | Format-Table FriendlyName, HealthStatus, OperationalStatus, Usage
Get-StorageJob
Get-VirtualDisk | Format-Table FriendlyName, HealthStatus, OperationalStatus
Get-NetIntentStatus | Format-List Host, IntentName, ConfigurationStatus, ProvisioningStatus
Get-SolutionUpdateEnvironment
```

Deployment validation failures

Symptom	Likely cause	Action
Validation fails at the diagnostic account or Key Vault audit logging	`Microsoft.Insights` resource provider not registered	Register it on the subscription and run Validate again
Environment Checker fails outbound connectivity	Firewall allow-list incomplete, or rules applied after Arc registration	Open the endpoints for your chosen topology; rerun the checker
Environment Checker fails ICMP to the gateway	Gateway blocks ICMP from the management pool	Allow ICMP from the management subnet to the default gateway
Deployment or Arc Resource Bridge behaves oddly with addresses	Node, cluster, DNS, proxy or pool addresses fall in `10.96.0.0/12` or `10.244.0.0/16`	Re-plan addresses outside the reserved Kubernetes ranges; these ranges cannot be changed
Registration or deployment rejects the local administrator	The built-in Administrator account was used	Create a separate local administrator, identical on every node
Deployment fails on node addressing	Nodes use DHCP under Local Identity	Set static addresses, gateway and DNS on every node
Name resolution fails	DNS zone missing, no host records for nodes and the system, or DNS cannot reach the Arc endpoints	Create the zone and A records; check forwarding; note that DNS servers cannot be changed after deployment
Cannot select a witness storage account	The account is already used by another system	Use a separate storage account for each system
Arc gateway option missing or cannot be added	The gateway must be chosen at deployment of a new instance on 2506 or later	Plan it before registration; it cannot be enabled afterwards
Management VLAN wrong after registration	VLAN not set on the physical adapters before registration	Management VLAN cannot be changed after deployment; set it before registering; redeploy if necessary

Symptom	Likely cause	Action
Machines show Not validated or not ready	OS versions or adapter configurations differ between nodes	Make all nodes identical; reregister
Portal access to nodes fails	SSH not enabled on the nodes	Enable SSH for Arc-enabled servers on each node

Day-2 platform

Symptom	Likely cause	Action
Insights shows a blank workbook	Configured under an hour ago, or the data collection rule lacks the required event channels or counters	Wait up to an hour, check the rule has the required channels and counters, then update Insights
Insights shows Needs update	The rule was changed, or a required counter or event was removed	Select Update in Insights to restore the required data sources
Update readiness reports Critical or Warning	A health check failed	Open the readiness details, follow the remediation link, then check again
Update stays Ready but never installs	Vendor validation of the new release is pending, or a readiness check fails	Confirm the release is available for your hardware; run `Start-SolutionUpdate -PrepareOnly` to see the checks
New feature release not offered	The hardware vendor has not yet validated it	Wait for the vendor sign-off; it can take a week or more
Policy shows non-compliant straight after assignment	Compliance evaluation is not instant	Start a compliance scan and wait for it to complete
Backup job fails	Source cannot reach the vault, or the protection route does not support the VM state	Check connectivity, then the support matrix of the backup solution you use: confirm that it protects Azure Local VMs at the host or guest level and which restore targets it offers (restoring to a different cluster brings the VM back unmanaged until it is re-registered)
Logical network VM has no IP	IP pool exhausted, or no static pool defined	Add an address range to the logical network
VM creation fails on a storage path	Not enough space at the storage path	Expand the volume or choose another path; never use the infrastructure volume

Faults

Symptom	Likely cause	Action
A drive shows Retired or Lost Communication	Failed media, or the drive was taken offline	Replace the drive if failed; wait for the repair job; confirm volumes `Healthy`
Replacement drive is not added to the pool	AutoPool disabled	Enable AutoPool, or add the disk to the pool manually
Volumes stay Degraded	A repair job is still running or cannot run	Check `Get-StorageJob`; ensure enough reserve capacity; do not stack other changes

Symptom	Likely cause	Action
An intent shows a configuration status other than Success	A setting Network ATC owns was changed, or an adapter failed	Review `Get-NetIntentStatus` and the Network ATC admin event log; fix the intent, not the adapter; restart Network ATC on each node in turn if a procedure requires it
A node is down	Power, hardware or network	Restore through the management path; when it returns allow storage to resync
Cluster is offline on a two-node system	A node and the witness are both lost	Restore one of them; verify the witness storage account and its connectivity
An Azure Local VM shows disconnected from Azure	Arc connectivity lost	Restore outbound connectivity; the VM reconnects inside the Arc agent's 45-day window

Replication and migration

Symptom	Likely cause	Action
Site Recovery replication unhealthy	Node cannot reach the service, or proxy TLS inspection blocks private endpoints	Check connectivity; bypass the proxy for the endpoint or disable inspection for it
Failed-back VM cannot be managed from Azure	It returned to a different cluster as an unmanaged VM	Register it on the new cluster and reconnect it to its Azure resource
Failover job cancelled and VM no longer replicates	A failover was cancelled in progress	Re-enable replication
Migrated VM has offline disks	SAN policy not set on the source	Bring disks online in the guest
Replication cannot start from VMware	BitLocker, shared disk, encrypted volume, or a missing VDDK package	Remove the blocker on the source; use a partner option if no VDDK is available

When to escalate

Collect and attach: the exact error text and time, the output of the first checks above, the readiness or health check details, the deployment correlation identifier, and the Azure Local version. Escalate when a repair job does not progress, when a node does not rejoin, when an update stops, or when a symptom needs a change you cannot reverse.